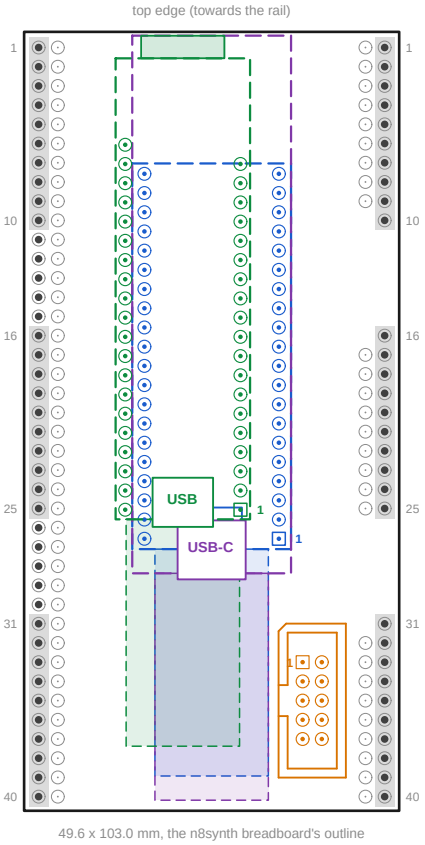
Checkpoint 2: 1:1 printout. Arrangement A, every module overlaid

The schematic's pin map. Outward face (the modules' side), seen from above. USB ends toward the board centre.



- Pico 2 holes, 2 x 20, rows 17.78 mm apart (square = pin 1)
- Pico 2 body, 21 x 51 mm
- ESP32-P4-WIFI6 body, 21 x 71 mm, in the Pico's holes
- Teensy 4.1 holes, 2 x 24, rows 15.24 mm apart
- Teensy 4.1 body, 17.8 x 61 mm
- USB plug keep-out, 15 x 30 mm per module: nothing taller than about 8 mm (tinted in each module's colour)
- J201 power header, 2 x 5 shrouded (KiCad outline)
- Deck hole (outer) and deck-cell spans 1-10, 16-25, 31-40
- Wiring hole (inner), joined to the deck hole beside it

Where things land (mm from the top-left corner)

Raspberry Pi Pico 2: rows x = 15.91 / 33.69; body y 17.4-68.4; keep-out to y 98.4
Waveshare ESP32-P4-WIFI6: rows x = 15.91 / 33.69; body y 0.5-71.6; keep-out to y 101.6
Teensy 4.1: rows x = 13.37 / 28.61; body y 3.5-64.5; keep-out to y 94.5
J201: pin 1 at (36.83, 83.35); shroud x 33.6-42.5, y 78.2-98.6

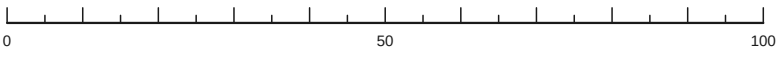
Why the Pico sits 17 mm down: an ESP32-P4 in the Pico's holes runs 16.9 mm past a Pico's far end. With the Pico at the top edge it would overhang the board, which ends a few mm from the rail. So the P4 ends 0.5 mm from the top edge, and its keep-out 1.4 mm from the bottom. The Teensy sits 3 mm lower: a card in its microSD socket stands just under 3 mm past its far end (measured), and now ends 0.5 mm in too.

Why the header is bottom right: every USB keep-out runs to the bottom centre, where the breadboard has its 2 x 8. Bottom right keeps it clear, puts the buck and rail filters in that corner, away from the converters, and in 6HP the right edge is the rear of the module, where the ribbon comes from. Pin 1 (top left, square) is -12 V, the red stripe.

No shared holes: the Teensy's left row is one pitch outside the Pico's, its right row two pitches inside. If both modules' headers are fitted, the Teensy's body sits over the Pico's left header (and the Pico over the Teensy's right one): fit one module's headers per board.

Check with real parts (page 2 has one module per outline):

1. Measure the 100 mm bar below.
2. Lay the n8synth breadboard on the outline: corners, then its edge-strip holes over the printed strip holes, positions 1 and 40 on both sides.
3. Lay each module face up with its USB end down, pin 1 (square) at the USB end: Pico 2 and the ESP32-P4 on the blue holes, Teensy 4.1 on the green. The P4's far end must not pass the top edge.
4. Hold each module at its real height (8.5 mm female header + the module) and plug in a USB cable: does the plug, and the cable's bend back into the case, stay inside its tinted keep-out?
5. Put the shrouded 2 x 5 header and the ribbon's plug on the orange outline. In 6HP the board stands at 90 degrees and the left strip meets the deck: is bottom right a good place for the ribbon?
6. Teensy: with a card in its microSD socket, does the card stay on the board? (First print: it stood just under 3 mm past the Teensy's far end, so the Teensy moved 3 mm down.)
7. Note anything that disagrees, in mm. Positions are in tools/analogue_board/placement.py.

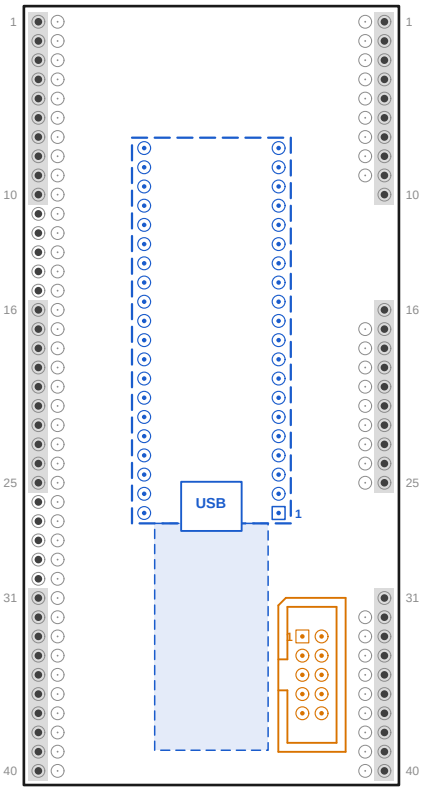


This bar must measure 100 mm. If not, reprint at 100% / Actual size, with fit-to-page and scaling turned off.

Checkpoint 2: one module per outline, to lay real parts on

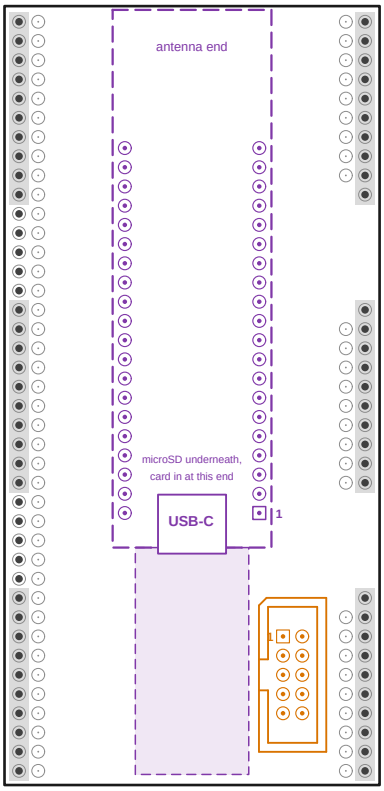
Same positions as page 1, without the overlap. Header and edge strips repeated on each.

Raspberry Pi Pico 2



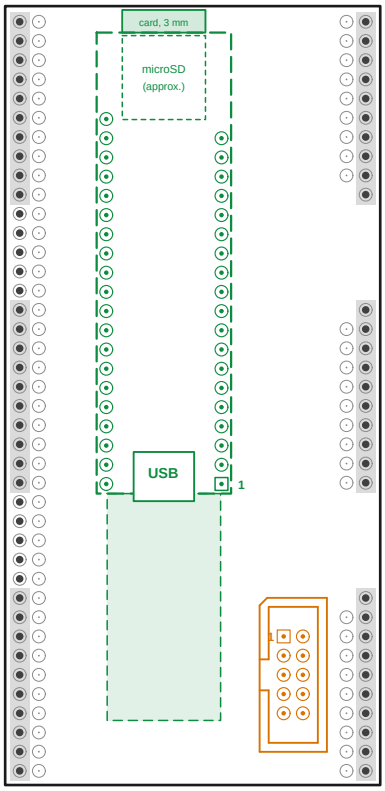
Body 21 x 51 mm, far end 17.4 mm down (room for the P4).
USB edge at 68.4 mm; keep-out to 98.4 mm.

Waveshare ESP32-P4-WIFI6



Body 21 x 71 mm in the Pico's holes. Far (antenna) end 0.5 mm from the top edge. USB-C edge at 71.6 mm; keep-out to 101.6 mm.
USB-C is serial only (CH343).
Bridge JP201 when fitted.

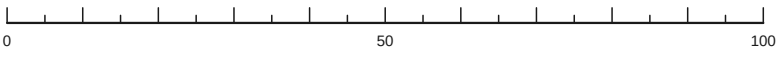
Teensy 4.1



Body 17.8 x 61 mm, far end 3.5 mm down, so a microSD card (3 mm) ends 0.5 mm from the top edge.
USB edge at 64.5 mm; keep-out to 94.5 mm.
Cut the VIN-VUSB pads first.

Laying the parts

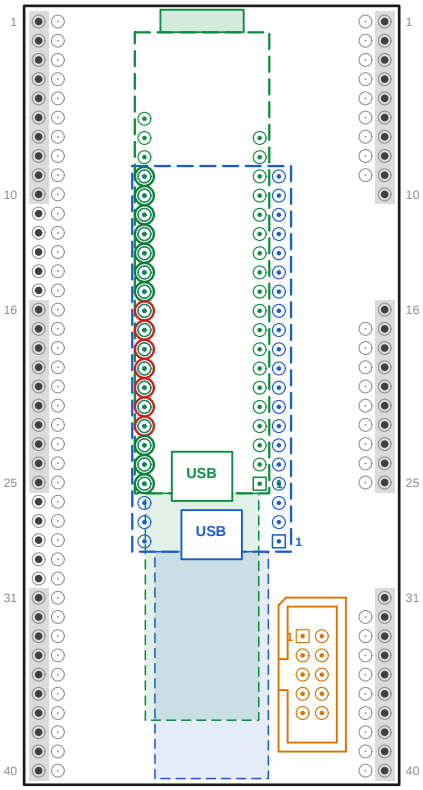
- Face up, USB end down, pin 1 (the square hole) at the USB end. The pins should sit on the dots.
- The keep-out is where the plug goes when the module is up on its headers; anything in it taller than about 8 mm stops the plug. Try your real cables: a right-angle plug may need less room.
- The P4's microSD card goes in at the USB end, under the board, and is meant to stay in.
- Edge-strip numbers are on the left-hand board; the grey bands are the deck cells.



This bar must measure 100 mm. If not, reprint at 100% / Actual size, with fit-to-page and scaling turned off.

Checkpoint 2: arrangement B (shared holes), for comparison only

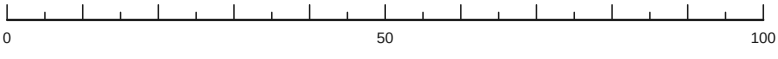
The Teensy's left row goes into the Pico's left-row holes. The schematic does NOT use this.



17 holes would be shared; 7 of them join two different nets in today's pin map (red rings). Green rings: same net, or a pin with no net.

Pico	name	net	Teensy	name	net
21	GP16	SPIA_MISO	32	D40	(no net)
22	GP17	CS_ADC	33	D41	(no net)
23	GND	GND	34	GND	GND
24	GP18	SPIA_SCK	35	D13	SPIA_SCK
25	GP19	SPIA_MOSI	36	D14	(no net)
26	GP20	CS_DAC	37	D15	(no net)
27	GP21	MCLK	38	D16	(no net)
28	GND	GND	39	D17	LED_DATA
29	GP22	I2S_DIN	40	D18	SDA
30	RUN	RUN	41	D19	SCL
31	GP26	BCK	42	D20	LRCK
32	GP27	LRCK	43	D21	BCK
33	AGND	GND	44	D22	EXP_INT
34	GP28	I2S_DOUT	45	D23	MCLK
35	ADC_VREF	(no net)	46	3V3	(no net)
36	3V3_OUT	(no net)	47	GND	GND
37	3V3_EN	(no net)	48	VIN	MOD_5V

Why A instead: a shared hole ties a Pico pin to whatever Teensy pin lands in it. The Pico needs all 26 of its GPIOs, so a GPIO hole that falls on a Teensy GND or 3.3 V pin is lost, and every other pin choice has to suit both modules at once. B saves holes and moves the Teensy 2.5 mm right; it would need the pin map redone. The table is computed from the current netlist.



This bar must measure 100 mm. If not, reprint at 100% / Actual size, with fit-to-page and scaling turned off.